Student academic course feedback application

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Abstract

- The Dream Destinations Hotel Booking Application is a travel-based project designed to help users explore various tourist locations and find suitable hotels based on their preferences. The application allows users to select destinations, enter booking details such as number of guests and rooms, and view available hotels with pricing information. It provides an interactive and user-friendly interface that makes searching and selecting accommodations easy and organized.

- The project uses structured design with HTML, CSS, and JavaScript to create dynamic hotel listings and calculate total booking costs based on user input. It demonstrates practical implementation of front-end development concepts including form handling, data storage using objects, and dynamic content generation. The application aims to provide a realistic hotel reservation experience while showcasing responsive design and interactive functionality suitable for academic submission.

Existing System

- Traditional hotel booking systems rely on **manual processes or basic search methods**, making it difficult to manage large amounts of booking data efficiently.
- Most basic systems **do not utilize efficient data structures**, which leads to slower search and booking operations.
- Managing room availability and booking requests without structured algorithms can result in **overbooking or inefficient allocation of rooms**.
- Users often need to **manually browse through multiple hotels and destinations**, which increases the time required to complete a booking.
- There is **limited support for waiting lists and booking prioritization**, which reduces system efficiency when rooms are fully booked.

Proposed System

- • The proposed **Dream Destinations Hotel Booking System** provides a digital platform for managing hotel reservations.
- The system consists of two major components:
- • **Web-based application (FWD)** developed using HTML, CSS, and JavaScript.
  • **Java-based DSA implementation** demonstrating algorithmic processing.
- Users can:
- • Signup / Login
  • Browse travel destinations
  • Select hotels
  • Book rooms with check-in and check-out dates.

Proposed System

- The system performs **room capacity checks and date-range availability validation**.
- A **waiting list mechanism** allows users to join when rooms are fully booked.
- The system suggests the **next available booking date** automatically.
- Booking cost is calculated using **number of rooms × number of nights**.
- All data is stored in **structured text files and browser storage** for efficient management.

```
+----------------------------+
|            USER            |
| Login / Signup             |
| Browse Destinations        |
| View Hotels                |
| Book Rooms                 |
+----------------------------+
              |
              v
+----------------------------+
|        WEB INTERFACE       |
| HTML Structure             |
| CSS Styling & Layout       |
| Responsive Design          |
| Navigation Links           |
+----------------------------+
              |
              v
+----------------------------+
|     JAVASCRIPT FUNCTIONALITY|
| Form Validation            |
| Dynamic Hotel Display      |
| Event Handling             |
| DOM Manipulation           |
+----------------------------+
              |
              v
+----------------------------+
|        BOOKING MODULE      |
| Select Destination         |
| Select Hotel               |
| Enter Check-in / Check-out |
| Calculate Booking Price    |
| Confirm Booking            |
+----------------------------+
              |
              v
+----------------------------+
|        DATA STORAGE        |
| Browser Storage            |
| (local Storage)             |
| Store Booking Details      |
| Store User Data            |
+----------------------------+
              |
              v
+----------------------------+
|           OUTPUT           |
| Booking Confirmation       |
| Hotel Availability Display |
| Payment Summary            |
+----------------------------+---
```

System Architecture

• The **Dream Destinations / Travel Mate Hotel Booking System** consists of two components:
Web-Based Interface (FWD) and **Java-based Data Processing System (DSA)**.

• The **Web Interface** built using **HTML, CSS, and JavaScript** allows users to **signup/login, browse destinations, view hotels, and enter booking details**.

• The **Java DSA system** manages booking operations using:

- **Linked List** – destination storage
- **Queue** – booking request processing
- **Stack** – booking history
- **Searching & Sorting** – hotel lookup and price ordering.

• The system performs **room availability checks, date validation, and waiting list management**.

• All system data (**users, destinations, hotels, bookings, waiting list**) is stored in **structured text files**.

Project Modules

- **1. Web Interface Module (FWD)**
  - Developed using **HTML, CSS, and JavaScript**.
  - Provides pages for **home, destinations, hotel listing, booking form, and payment**.
- **2. Authentication Module**
  - Handles **user signup and login validation**.
  - Manages user sessions using **browser storage (localStorage)**.
- **3. Booking Management Module**
  - Allows users to **select hotels, enter booking dates, and confirm reservations**.
  - Calculates booking cost based on **rooms and stay duration**.

Project Modules

- **4. Availability & Waiting List Module**
  - Checks **room availability and date conflicts**.
  - Manages **waiting list when hotels are fully booked**.
- **5. Data Storage Module**
  - Stores data in **text files and browser storage**.
  - Maintains records of **users, hotels, bookings, and waiting list**.
- **6. DSA Processing Module**
  - Implements **Linked List, Queue, Stack, Searching, and Sorting algorithms**.
  - Ensures efficient processing of booking operations.

Objectives

- To design and develop a **hotel booking platform** using web technologies and Java.
- To provide a **user-friendly interface** for browsing destinations, viewing hotels, and making bookings.
- To implement **Data Structures and Algorithms** such as Linked List, Queue, Stack, Searching, and Sorting in a practical application.
- To manage **hotel availability, booking dates, and waiting lists** efficiently.
- To demonstrate **data storage and retrieval using structured text files and browser storage**.
- To integrate **frontend web development with algorithmic data processing** in a real-world system.

Co’s Attainments - FWD

CO1 Attainment

Statement: Apply Internet fundamentals, HTML structure, and CSS styling to design structured web pages.

CO1 Topic	Implementation in Hotel Booking Web System
HTML Document Structure	Used to create pages such as Home, Destinations, Hotels, Booking
Navigation Links	Connect pages like destinations → hotels → booking
CSS Styling	Applied for layout design, fonts, and responsive UI
Web Concepts	Browser-based hotel booking interface

Implementation Scenario

- Developed multiple web pages such as **Home, Destinations, Booking, Hotels, Payment, and Confirmation**.
- Used **HTML document structure and hyperlinks** to connect pages.
- Applied **CSS styling for layout, colors, and responsive design**.

Co’s Attainments - FWD

CO2 Attainment

Statement: Apply HTML forms, semantic tags, and comprehensive CSS layouts.

CO2 Topic	Implementation
HTML Forms	Booking form with input fields
Semantic Tags	Structured page layout
CSS Layouts	Flexbox/grid-based layouts
Responsive Design	Mobile-friendly interface

Implementation Scenario

- Designed **hotel booking forms for entering booking details**.
- Used **semantic HTML elements and structured layout**.
- Implemented **responsive design using CSS layouts**.

Co’s Attainments - FWD

CO3 Attainment

Statement: Apply JavaScript programming essentials.

CO3 Topic	Implementation
JavaScript Logic	Booking cost calculation
Conditions & Loops	Input validation and form checks
Functions	Booking and payment processing
Arrays & Objects	Store hotel and booking data

Implementation Scenario

- Implemented **JavaScript logic for booking calculations**.
- Used **functions, conditions, and loops for validating user input**.
- Managed booking information using **arrays and objects**.

Co’s Attainments - FWD

CO4 Attainment

Statement: Apply JavaScript interactivity and DOM manipulation.

CO4 Topic	Implementation
DOM Manipulation	Dynamic hotel display
Event Handling	Booking and form events
Browser Storage	localStorage for booking data
Async Handling	JavaScript-based UI updates

Implementation Scenario

- Implemented **DOM manipulation to dynamically display hotels and destinations**.
- Used **event handling for form submission and booking actions**.
- Stored booking data using **browser localStorage**.

Co’s Attainments - FWD

CO5 Attainment

Statement: Apply advanced web development and deployment concepts.

CO5 Topic	Implementation
Form Validation	Booking form validation
Performance	Efficient page navigation
SEO Concepts	Structured HTML pages
Deployment	Ready-to-host web application

Implementation Scenario

- Implemented **form validation and user input handling**.
- Optimized web pages for **better loading and usability**.
- Designed the system to be **deployable as a multi-page web application**.

Co’s Attainments - DSA

CO1 Attainment

Statement: Analyze algorithms and implement searching and sorting techniques.

CO1 Topic	Implementation
Linear Search	Search hotels by name
Binary Search	Efficient hotel lookup
Bubble Sort	Sort hotels by price
Algorithm Analysis	Compare search efficiency

Implementation Scenario

- Implemented **Linear Search and Binary Search for hotel lookup**.
- Implemented **Bubble Sort to sort hotels based on price**.
- Compared search and sort efficiency for booking operations.

Co’s Attainments - DSA

CO2 Attainment

Statement: Design and implement Abstract Data Types using linked lists.

CO2 Topic	Implementation
Linked List ADT	Destination storage
Insert Operation	Add new destinations
Traverse Operation	Display destinations
Search Operation	Find destination

- **Implementation Scenario**

• Implemented **Linked List to store travel destinations**.
• Performed **insert, traverse, and search operations** on destinations.

Co’s Attainments - DSA

CO3 Attainment

Statement: Apply stacks and queues to real-world workflows.

CO3 Topic	Implementation
Stack	Booking history management
Queue	Booking request processing
Circular Queue Concept	Request handling
Priority Processing	Waiting list handling

- **Implementation Scenario**

• Used **Queue to manage booking requests in FIFO order**.
• Used **Stack to maintain booking history for undo operations**.

Co’s Attainments - DSA

CO4 Attainment

Statement: Implement hash-based data structures for efficient lookup.

CO4 Topic	Implementation
Hash Function	Email-based lookup
Hash Table	User authentication
Collision Handling	Efficient lookup management
Map Structures	User data access

- **Implementation Scenario**

• Implemented **Hash Table for fast user login lookup using email**.
• Used hashing to improve **user authentication performance**.

Co’s Attainments - DSA

CO5 Attainment

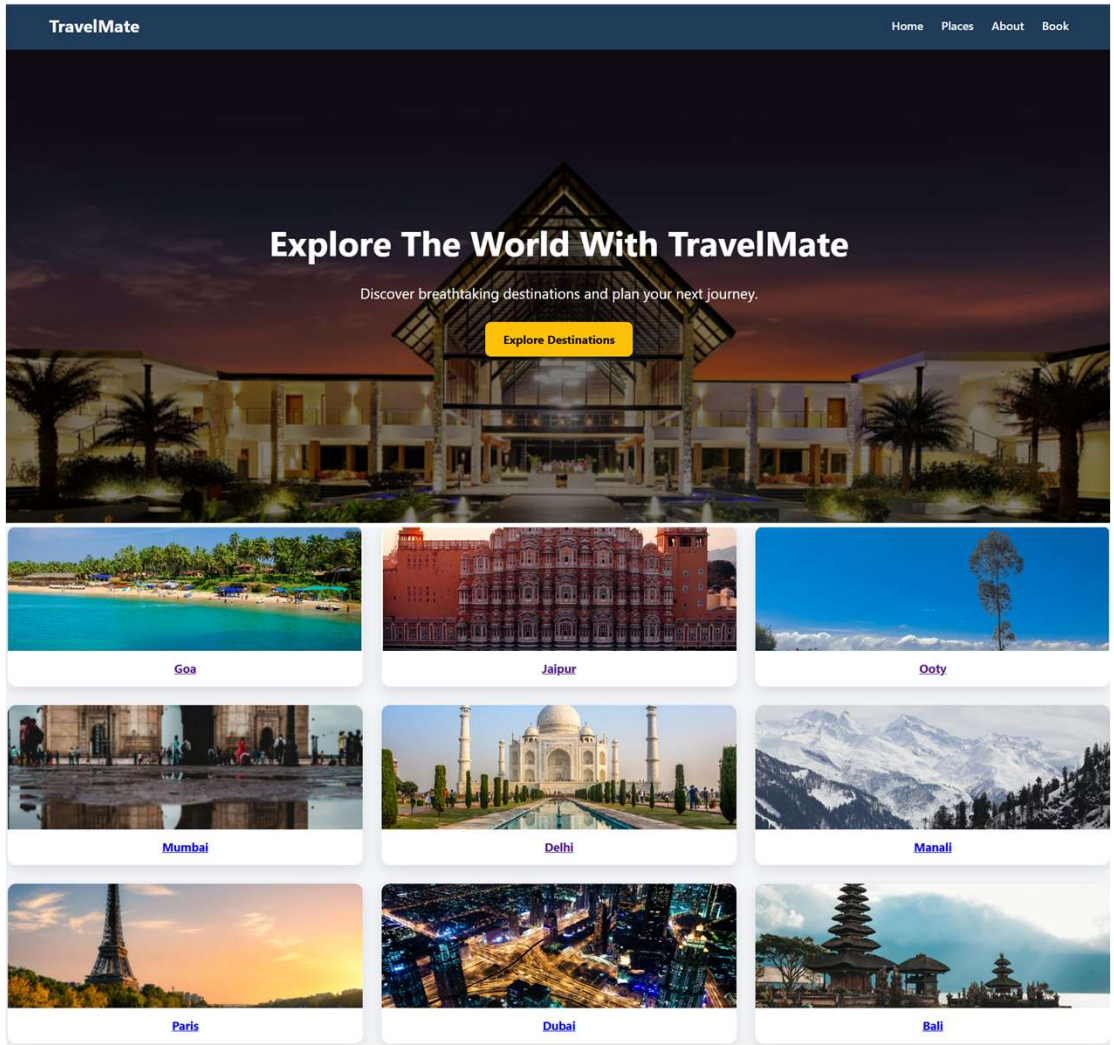
Statement: Develop real-world applications using linear data structures.

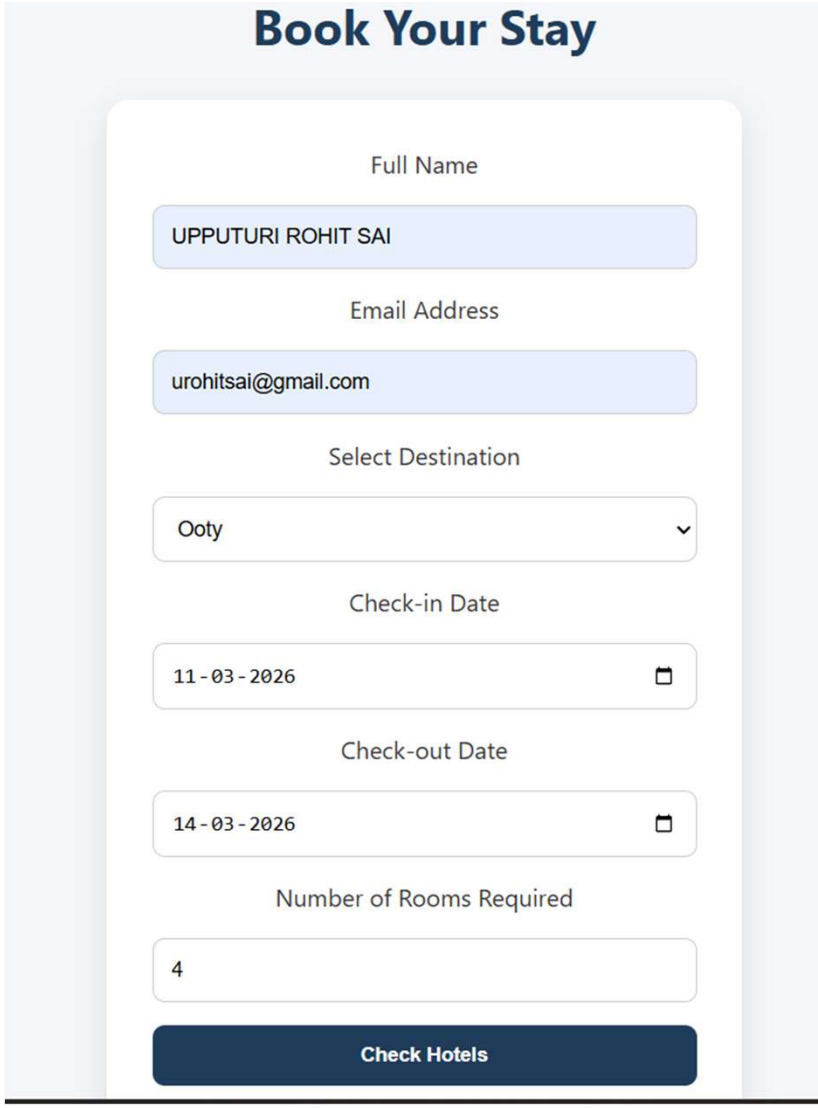
CO5 Topic	Implementation
Linear Data Structures	Linked List, Stack, Queue
Real-world Application	Hotel booking management
Data Processing	Booking and availability logic
File Storage	users.txt, hotels.txt, bookings.txt

- **Implementation Scenario**

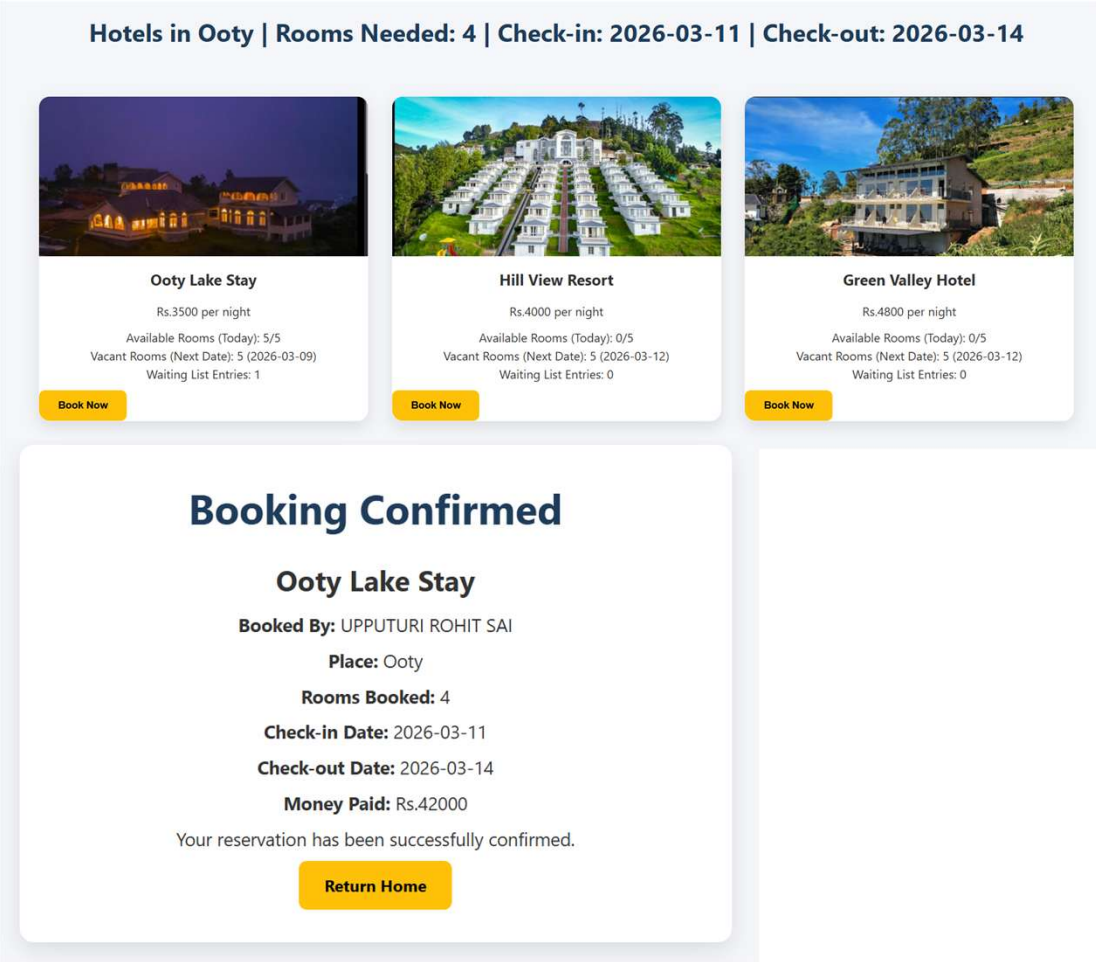
• Built a **complete hotel booking system using DSA concepts**.
• Applied **Linked List, Stack, Queue, Searching, and Sorting** in a real application.
• Managed **bookings, waiting lists, and hotel data efficiently**.

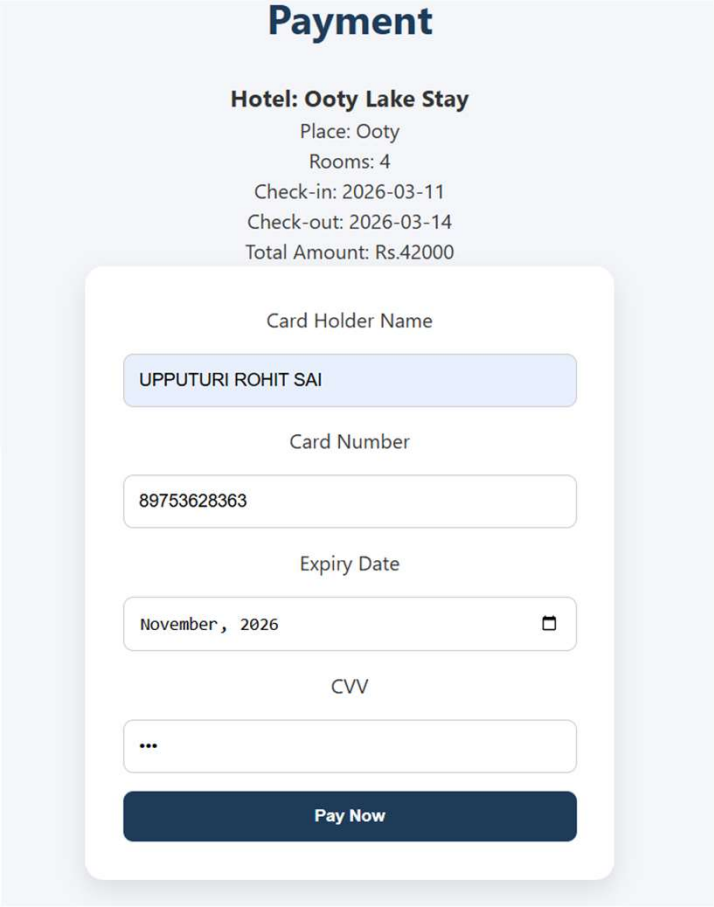
Screenshots for FWD





Screenshots for FWD





Screenshots for DSA

```
===== Dream Destinations Hotel Booking System =====

1. Login
2. Signup
3. Exit
Enter choice: 1
Enter email: urohitsai@gmail.com
Enter password: HIT
Login successful. Welcome, Rohit Sai.

===== BOOKING FLOW =====

Available Destinations:
1. Goa
2. Manali
3. Delhi
4. Ooty
5. Jaipur
6. Mumbai
7. Dubai
8. Bali
9. Paris
Choose destination ID: 8

Hotels in selected destination:
HotelID: 24, DestinationID: 8, Name: Ubud Green Stay, Price: 6800.0, Available Rooms: 5, Vacant Rooms: 5 (2026-03-09)
HotelID: 23, DestinationID: 8, Name: Island Paradise Hotel, Price: 7500.0, Available Rooms: 5, Vacant Rooms: 5 (2026-03-09)
HotelID: 22, DestinationID: 8, Name: Bali Beach Resort, Price: 9000.0, Available Rooms: 5, Vacant Rooms: 5 (2026-03-09)
Choose hotel ID: 23
Enter rooms to book: 3
Enter check-in date (YYYY-MM-DD): 2026-03-11
Enter check-out date (YYYY-MM-DD): 2026-03-14
Enter name: UPPUTURI ROHIT SAI
Enter email: urohitsai@gmail.com
```

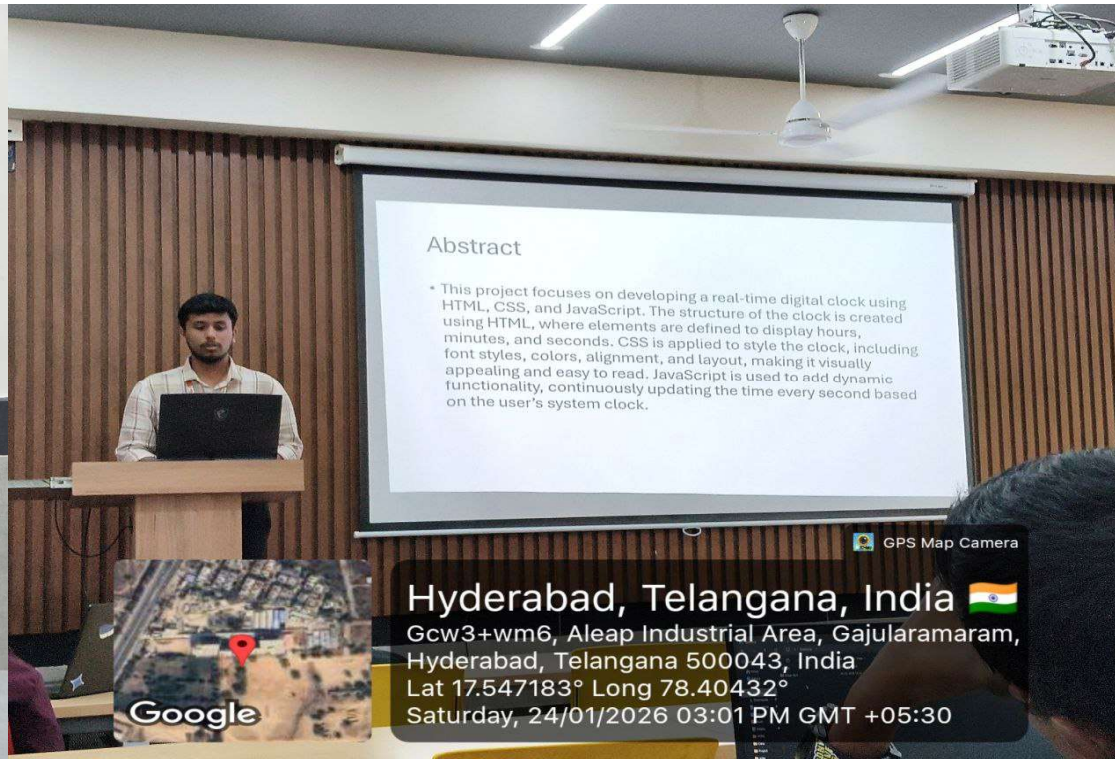
```
===== PAYMENT =====
Card holder name: UPPUTURI ROHIT SAI
Card number: 789764536789
Expiry (MM/YY): 09/27
CVV: 789

===== BOOKING CONFIRMATION =====
Booking ID: 3
Booked By: UPPUTURI ROHIT SAI
Email: urohitsai@gmail.com
Place: Bali
Hotel: Island Paradise Hotel
Check-in Date: 2026-03-11
Check-out Date: 2026-03-14
Rooms: 3
Nights: 3
Total Money: 67500.0
Status: CONFIRMED
Returning to Home Page...
```

Conclusion

- The **Dream Destinations / Travel Mate Hotel Booking System** successfully integrates **web development and data structure concepts** to create a functional travel booking platform.
- The **web version (FWD)** provides a user-friendly interface for browsing destinations, selecting hotels, and completing bookings.
- The **Java-based implementation (DSA)** demonstrates practical use of **Linked List, Queue, Stack, Searching, and Sorting algorithms**.
- The system efficiently manages **room availability, booking requests, waiting lists, and pricing calculations**.
- This project shows how **frontend technologies and algorithmic data processing can be combined to build real-world applications**.

Geo- Tag Photos



THANK YOU